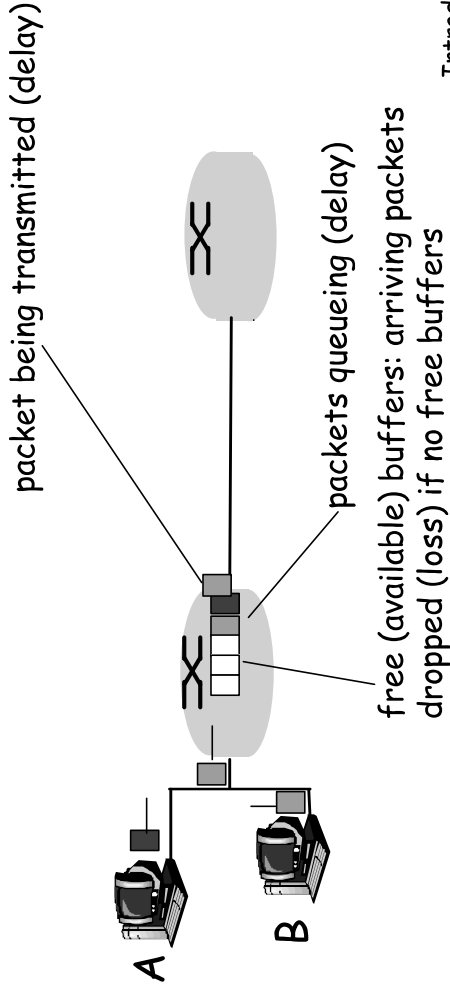


How do loss and delay occur?

packets *queue* in router buffers

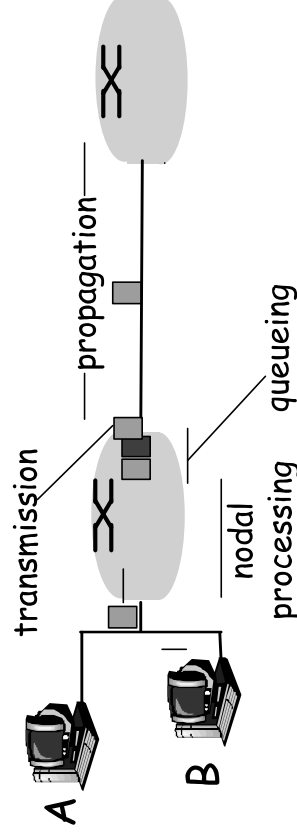
- ✍ packet arrival rate to link exceeds output link capacity
- ✍ packets queue, wait for turn



Introduction 1-50

Four sources of packet delay

- ✍ 1. nodal processing:
 - ✍ check bit errors
 - ✍ determine output link
- ✍ 2. queuing
 - ✍ time waiting at output link for transmission
 - ✍ depends on congestion level of router



Introduction 1-51

Delay in packet-switched networks

3. Transmission delay:

✂ R=link bandwidth (bps)

✂ L=packet length (bits)

✂ time to send bits into

$$\text{link} = L/R$$

4. Propagation delay:

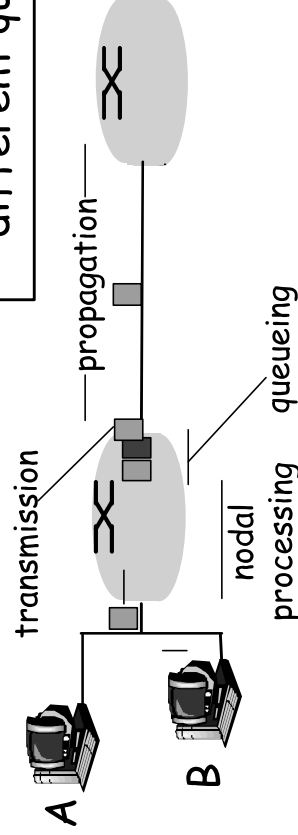
✂ d = length of physical link

✂ s = propagation speed in

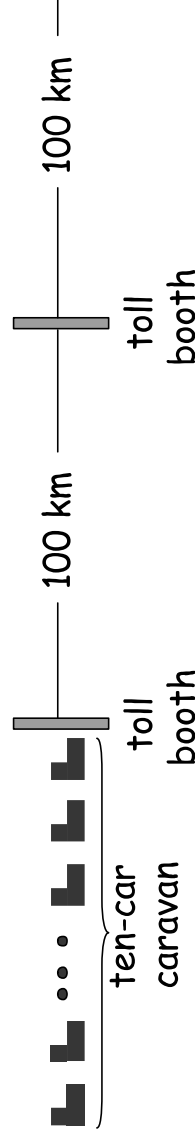
medium ($\sim 2 \times 10^8$ m/sec)

✂ propagation delay = d/s

Note: s and L/R are very different quantities!



Caravan analogy



✂ Cars "propagate" at 100 km/hr

✂ Toll booth takes 12 sec to service a car (transmission time)

✂ car~bit; caravan ~ packet

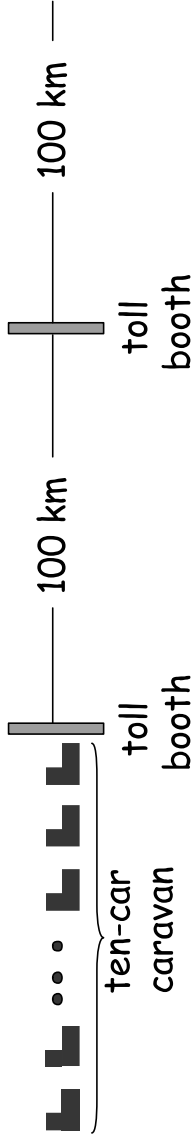
✂ Q: How long until caravan is lined up before 2nd toll booth?

✂ Time to "push" entire caravan through toll booth onto highway = $12 \times 10 = 120$ sec

✂ Time for last car to propagate from 1st to 2nd toll booth: $100\text{km}/(100\text{km/hr}) = 1$ hr

✂ A: 62 minutes

Caravan analogy (more)



- ✂ Cars now “propagate” at 1000 km/hr
- ✂ Toll booth now takes 1 min to service a car
- ✂ Q: Will cars arrive to 2nd booth before all cars serviced at 1st booth?

✂ Yes! After 7 min, 1st car at 2nd booth and 3 cars still at 1st booth.

- ✂ 1st bit of packet can arrive at 2nd router before packet is fully transmitted at 1st router!
- ✂ See Ethernet applet at AWL Web site

Nodal delay

$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$

- ✂ d_{proc} = processing delay
 - ✂ typically a few microsecs or less
- ✂ d_{queue} = queuing delay
 - ✂ depends on congestion
- ✂ d_{trans} = transmission delay
 - ✂ $= L/R$, significant for low-speed links
- ✂ d_{prop} = propagation delay
 - ✂ a few microsecs to hundreds of msecs